

## **Class Assignment on DRL**

### **Q1. Real-World Case of Priority Experience Replay in DRL**

Imagine training a robot to navigate a maze. The robot has to learn the correct policy by interacting with the environment. Most of the experiences might be unimportant (e.g., the robot moving around without getting closer to the goal), but occasionally, it may take a crucial action that brings it close to the goal.

- With uniform sampling, these critical experiences might not be replayed frequently enough, slowing down learning.
- With PER, experiences where the robot makes large mistakes or rare progress toward the goal are replayed more often, accelerating the learning process.

### **Q2. DQN - Autonomous Traffic Light Control System**

**Problem Statement:** In smart cities, managing traffic efficiently is crucial. Traffic lights need to adapt dynamically based on real-time traffic conditions to minimize congestion and travel time. Traditional, rule-based systems are inefficient because they do not adapt to changes in traffic patterns.

**Solution with DQN:**

- **The state of the agent:** The traffic situation at an intersection, such as the number of cars in each lane and waiting times.
- **Actions:** Possible states of traffic lights (e.g., green for one direction, red for others, or yellow lights).
- **Reward:** A reward is given for reducing vehicle waiting time and congestion at the intersection.

Using DQN, the system learns by interacting with traffic environments (e.g., simulations) to minimize waiting time at intersections.

**Limitation:** DQN suffers from overestimation bias in complex traffic patterns, sometimes choosing non-optimal policies due to noisy Q-values.

### **Q3. Double DQN - Stock Trading Bot**

**Problem Statement:** In financial markets, algorithmic trading bots are used to buy and sell stocks at the right times to maximize profits. The environment is highly volatile, with noisy market data, making it essential to make accurate predictions about future stock movements.

**Solution with Double DQN:**

- **State:** Market indicators like current stock price, moving averages, and trading volume.
- **Actions:** Buy, sell, or hold a stock.
- **Reward:** Profit or loss from taking a particular action.

Why Double DQN? The standard DQN algorithm may overestimate Q-values in this noisy environment, leading to poor trading decisions. By decoupling action selection (via the online network) and action evaluation (via the target network), Double DQN ensures more stable and accurate Q-value estimates, preventing the bot from making overly optimistic trades.

#### **Q4. Dueling DQN - Autonomous Drone Navigation**

**Problem Statement:** In autonomous drones, effective navigation in unknown environments (like urban areas or forests) is critical. The drone needs to learn to avoid obstacles and find the shortest path to a target, like a delivery location or a search-and-rescue target.

##### **Solution with Dueling DQN:**

- State: The current location, distance to obstacles, and distance to the goal.
- Actions: Move forward, turn left, turn right, or hover.
- Reward: A positive reward for moving closer to the target and a penalty for collisions.

##### **Why Dueling DQN?**

- In many states, actions like moving slightly left or right might not matter much (e.g., when the drone is far from obstacles).
- The dueling architecture allows the network to separate state value from the action advantage, meaning the drone learns how good it is to be in a particular state even when the actions don't have an immediate impact.
- This makes the learning process more efficient and helps the drone navigate in real time without unnecessary computation.

#### **Q5. DQN – Robot Path Planning in Warehouses**

##### **Problem Statement:**

Autonomous robots are used in e-commerce warehouses (e.g., Amazon) to pick and transport products. The robots need to find the shortest and safest paths to shelves and avoid collisions with other robots or obstacles.

- State: Position of the robot, nearby obstacles, target location.
- Actions: Move forward, turn left, turn right, or stop.
- Reward: Positive reward for reaching the target quickly and a penalty for collisions or taking too long.

#### **Q6. Double DQN – Autonomous Taxi Dispatch System**

##### **Problem Statement:**

In ride-hailing services like Uber or Ola, the goal is to efficiently assign taxis to customer requests to minimize waiting times and maximize profits. This involves predicting customer demand, driver availability, and optimizing the dispatch process.

- State: Location of available taxis, location of customers, and time of day.
- Actions: Assign a particular taxi or delay dispatch.
- Reward: Positive reward for minimizing waiting time and maximizing profit.

### **Q7. Dueling DQN – Online Video Streaming Quality Optimization**

Problem Statement:

In streaming platforms like Netflix or YouTube, the system must dynamically adjust video quality based on network conditions (bandwidth fluctuations) to provide the best user experience without buffering.

- State: Current bandwidth, buffer size, and video bitrate.
- Actions: Increase, decrease, or maintain video quality.
- Reward: Positive reward for improving video quality without buffering.

### **Q8. DQN – Energy Management in Smart Grids**

Problem Statement:

Smart grids manage energy distribution by balancing supply and demand in real-time. The system must decide how much energy to store, release, or purchase from external sources based on fluctuating demands and energy costs.

- State: Energy demand, supply, storage levels, and market prices.
- Actions: Store, release, or buy energy.
- Reward: Positive reward for meeting demand at minimal cost.

### **Q9. Double DQN – Self-Driving Car Lane Changing**

Problem Statement:

Autonomous vehicles need to make lane-changing decisions safely and efficiently, considering traffic patterns, vehicle speeds, and road conditions.

- State: Vehicle's speed, distance to other vehicles, and available lanes.
- Actions: Change lane left, change lane right, or stay in the current lane.
- Reward: Positive reward for smooth and safe lane changes, penalties for collisions or abrupt stops.

### **Q10. Dueling DQN – Personalized Advertising Recommendation System**

Problem Statement:

E-commerce platforms need to recommend personalized advertisements based on user behavior to maximize click-through rates (CTR) and conversions.

- State: User's browsing history, demographics, and past interactions.
- Actions: Recommend specific ads or no ads at all.
- Reward: Positive reward for clicks or conversions, penalties for showing irrelevant ads.